SIMATS ENGINEERING

ASSESSMENT TOOL I — Uncertainty Visualization Lab

Course: Data Handling and Visualization	**Code: DSA06**	**CO: CO5**
Duration: 180 min	**Total Marks: 100**	**Reg No :192524108** **Name: S. Lakshita**

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Uncertainty Visualization Lab

Weightage: 20%

Code of Conduct:

I S.LAKSHITA(192524108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Understanding of Uncertainty 25%	Visualization of Uncertainty 25%	Application of Visualization Principles 20%	Organization 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Uncertainty Visualization Lab – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualization, interpret uncertainty, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Bootstrapping and Confidence Intervals – Point Estimate

A quality-control team records the following delivery times (in minutes) for 12 randomly selected orders. Using R, calculate the sample mean and obtain a 95% bootstrap confidence interval for the mean using at least 5,000 bootstrap resamples. Visualize the bootstrap distribution and mark the observed mean.

Data: 31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36

Expected R tasks: resampling, point estimate, confidence interval, histogram/density plot,

interpretation.

Q2. Uncertainty in Trend Estimation

The following data represent monthly advertising expenditure and sales. Fit a linear regression model in R to estimate the sales trend. Plot the observations with the fitted regression line and a 95% confidence band. Explain how the uncertainty changes across the range of advertising expenditure.

Data: Advertising = 10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40;

Sales = 48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84

Expected R tasks: lm(), prediction interval/confidence interval, ggplot2 visualization, trend

interpretation.

Q3. Overplotting, Jittering and Transparency

A survey records the number of hours studied and examination scores for 30 students. Several students have identical values. Create a scatterplot in R first without modification, then redesign it using partial transparency and jittering. Compare the two plots and explain how overplotting affects interpretation.

Data: Hours = 2,3,3,4,4,4,5,5,5,5,6,6,6,6,7,7,7,7,7,8,8,8,8,8,9,9,9,10,10;

Scores = 45,52,52,58,60,60,64,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94

Expected R tasks: geom_point(), alpha, position_jitter(), comparison of overplotting.

Q4. 2D Histogram and Density Estimation

A researcher measures height and weight for 20 participants. Construct a 2D histogram and a 2D density visualization in R. Use the plots to identify regions with high observation density. State which visualization communicates concentration more clearly and justify your choice.

Data: Height = 150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186;
Weight = 48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88

Expected R tasks: 2D binning, density estimation, contour/density layers, interpretation of

concentration.

Q5. Color Scales, Accessibility and Misleading Visualization

The following category-wise satisfaction scores must be visualized in R. Create two plots: (1) a plot using a nonmonotonic/diverging color scheme that could mislead interpretation, and (2) a redesigned plot using an ordered, color-vision-deficiency-friendly sequential scale. Explain why the second design is more effective and identify the visualization pitfall in the first design.

Data: Category = A,B,C,D,E,F,G;

Satisfaction = 42,48,55,61,67,74,82

Expected R tasks: color mapping, scale selection, accessibility-aware redesign, critique of misleading color use.

Suggested R packages: ggplot2, dplyr, boot (or base R).

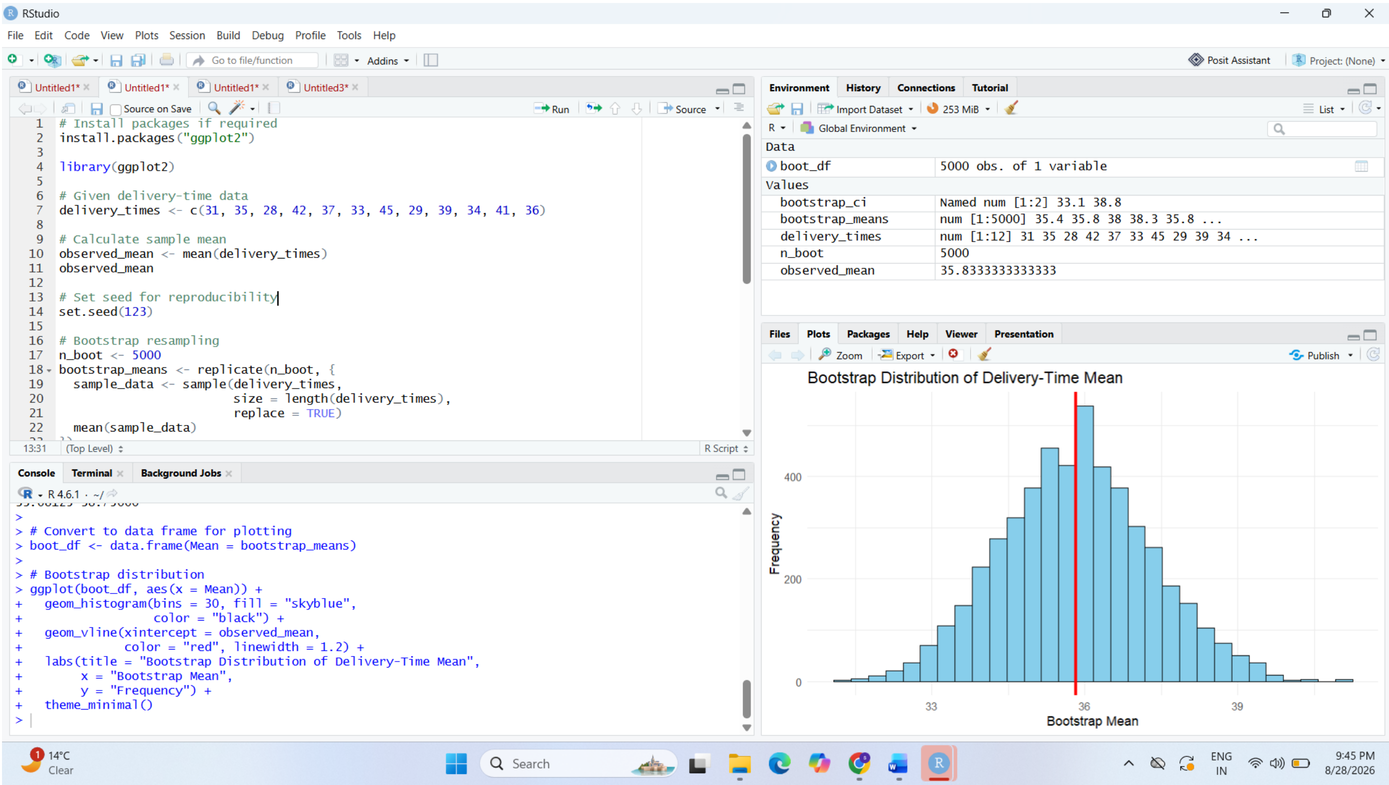
Assessment focus: Correct R code, appropriate uncertainty visualization, interpretation, and identification of misleading design choices.

1) Bootstrapping and Confidence Intervals – Point Estimate

```
# Install packages if required
install.packages("ggplot2")

library(ggplot2)

# Given delivery-time data
delivery_times <- c(31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36)

# Calculate sample mean
observed_mean <- mean(delivery_times)
observed_mean

# Set seed for reproducibility
set.seed(123)

# Bootstrap resampling
n_boot <- 5000
bootstrap_means <- replicate(n_boot, {
  sample_data <- sample(delivery_times,
                        size = length(delivery_times),
                        replace = TRUE)
  mean(sample_data)
})
```

```r
# Calculate 95% bootstrap confidence interval
bootstrap_ci <- quantile(bootstrap_means,
                         probs = c(0.025, 0.975))

bootstrap_ci

# Convert to data frame for plotting
boot_df <- data.frame(Mean = bootstrap_means)

# Bootstrap distribution
ggplot(boot_df, aes(x = Mean)) +
  geom_histogram(bins = 30, fill = "skyblue",
                 color = "black") +
  geom_vline(xintercept = observed_mean,
             color = "red", linewidth = 1.2) +
  labs(title = "Bootstrap Distribution of Delivery-Time Mean",
       x = "Bootstrap Mean",
       y = "Frequency") +
  theme_minimal()
```

Interpretation:
The sample mean represents the observed average delivery time. The bootstrap distribution shows the variability of the estimated mean across 5,000 resamples. The 95% bootstrap confidence interval gives a range of plausible values for the population mean. The red vertical line represents the observed sample mean.



2) Uncertainty in Trend Estimation

The assessment asks for a linear regression model, fitted regression line, and a **95% confidence band**.

Install packages if required

install.packages("ggplot2")

library(ggplot2)

Given data

Advertising <- c(10, 12, 15, 18, 20, 23,

25, 28, 30, 33, 36, 40)

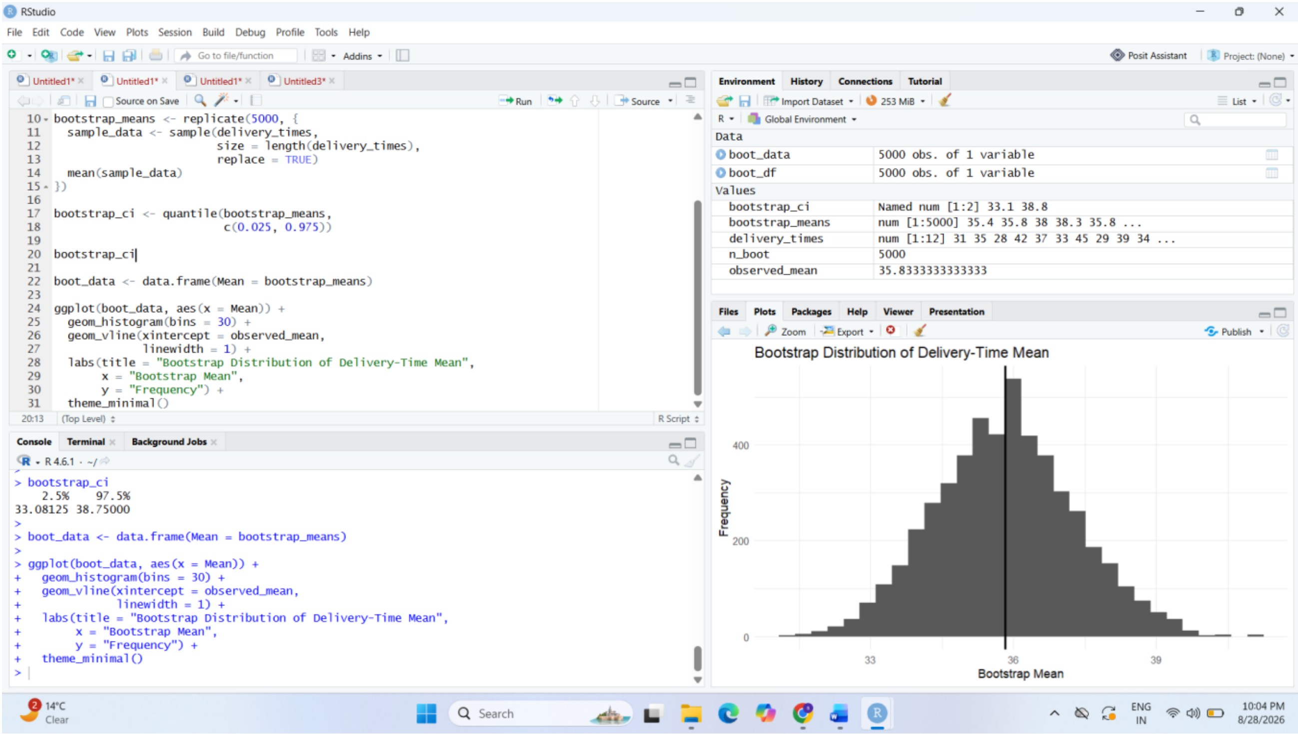
Sales <- c(48, 51, 53, 57, 60, 61,

66, 69, 72, 75, 79, 84)

```r
# Create data frame
data <- data.frame(Advertising, Sales)

# Fit linear regression model
model <- lm(Sales ~ Advertising, data = data)

# Display model summary
summary(model)

# Create prediction data
new_data <- data.frame(
  Advertising = seq(min(Advertising),
                    max(Advertising),
                    length.out = 100)
)

# Calculate 95% confidence interval
prediction <- predict(model,
                      newdata = new_data,
                      interval = "confidence",
                      level = 0.95)

new_data$fit <- prediction[, "fit"]
new_data$lwr <- prediction[, "lwr"]
new_data$upr <- prediction[, "upr"]
```

```
# Plot observations, regression line and confidence band

ggplot(data, aes(x = Advertising, y = Sales)) +

 geom_point(size = 3) +

 geom_ribbon(data = new_data,

       aes(x = Advertising,

         ymin = lwr,

         ymax = upr),

       alpha = 0.2,

       inherit.aes = FALSE) +

 geom_line(data = new_data,

      aes(x = Advertising, y = fit),

      linewidth = 1) +

 labs(title = "Advertising Expenditure vs Sales",

   x = "Advertising Expenditure",

   y = "Sales") +

 theme_minimal()
```

Interpretation:
The regression line shows a positive relationship between advertising expenditure and sales. As advertising expenditure increases, sales generally increase. The shaded region represents the 95% confidence band. The uncertainty is generally smaller near the center of the observed advertising range and increases toward the boundaries because there are fewer observations supporting the fitted trend there.



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3) Overplotting, Jittering and Transparency

The question specifically requires comparing the original scatterplot with redesigned versions using **transparency and jittering**.

Install package if required

install.packages("ggplot2")

library(ggplot2)

Given data

Hours <- c(2,3,3,4,4,4,5,5,5,5,

6,6,6,6,6,7,7,7,7,7,

8,8,8,8,8,9,9,9,10,10)

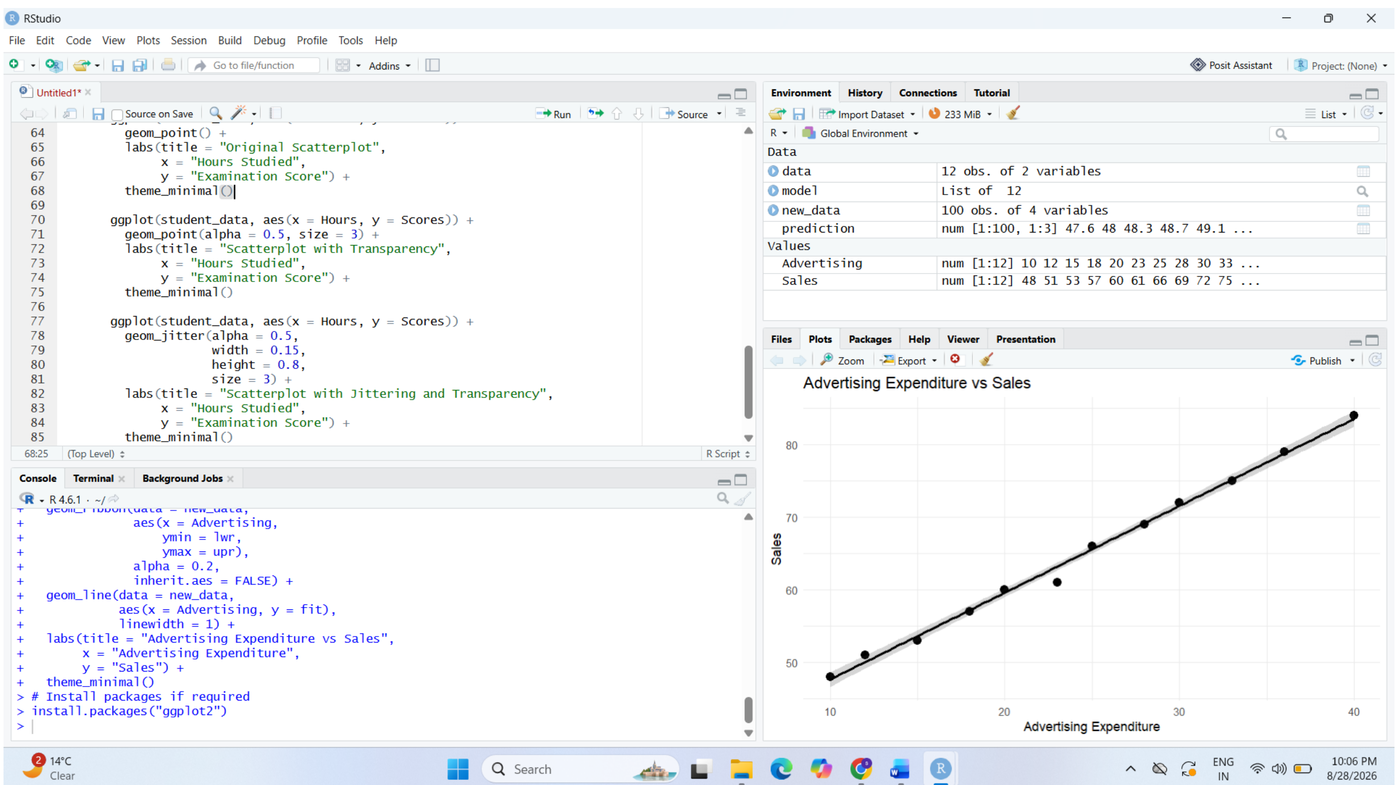
Scores <- c(45,52,52,58,60,60,64,66,66,66,

```
      70,72,72,72,72,76,78,78,80,80,
      84,84,85,85,85,88,90,90,94,94)

student_data <- data.frame(Hours, Scores)

# Original scatterplot
ggplot(student_data, aes(x = Hours, y = Scores)) +
  geom_point() +
  labs(title = "Original Scatterplot",
       x = "Hours Studied",
       y = "Examination Score") +
  theme_minimal()

# Scatterplot with transparency
ggplot(student_data, aes(x = Hours, y = Scores)) +
  geom_point(alpha = 0.5, size = 3) +
  labs(title = "Scatterplot with Transparency",
       x = "Hours Studied",
       y = "Examination Score") +
  theme_minimal()

# Scatterplot with jittering and transparency
ggplot(student_data, aes(x = Hours, y = Scores)) +
  geom_jitter(alpha = 0.5,
              width = 0.15,
              height = 0.8,
```

```
      size = 3) +
 labs(title = "Scatterplot with Jittering and Transparency",
    x = "Hours Studied",
    y = "Examination Score") +
 theme_minimal()
```

Interpretation:
The original scatterplot can hide overlapping observations because several students have identical study hours and scores. Transparency makes dense regions more visible because overlapping points become darker. Jittering slightly separates overlapping points, allowing individual observations to be seen more clearly. Therefore, the redesigned visualization provides a better representation of the distribution of students.

Visualization pitfall:
Overplotting can make multiple observations appear as a single point, causing the frequency of observations to be underestimated.



4) 2D Histogram and Density Estimation

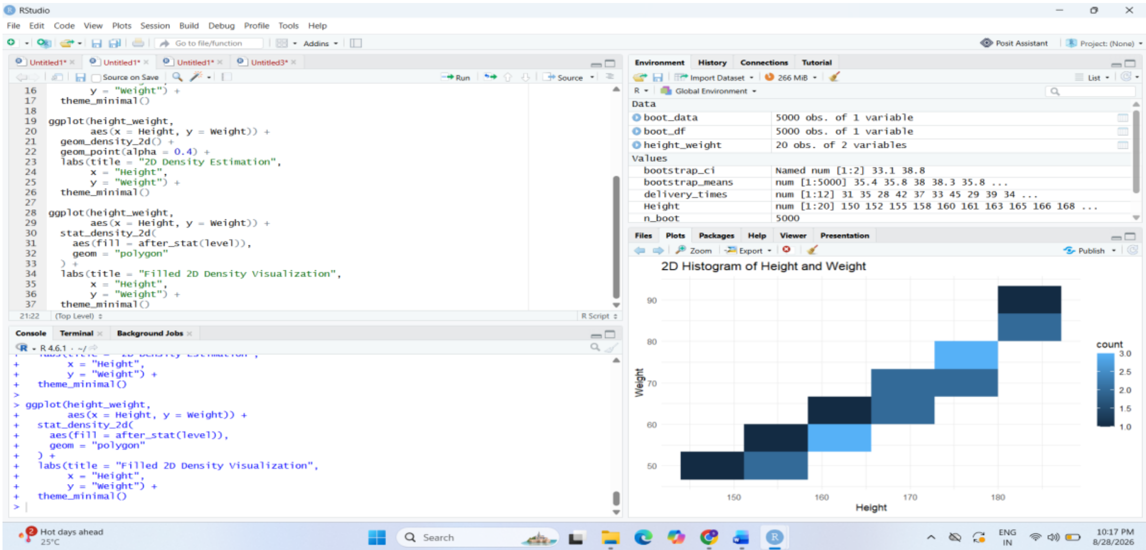
The assessment requires both **2D binning and density estimation**, followed by a comparison of which visualization communicates concentration more clearly.

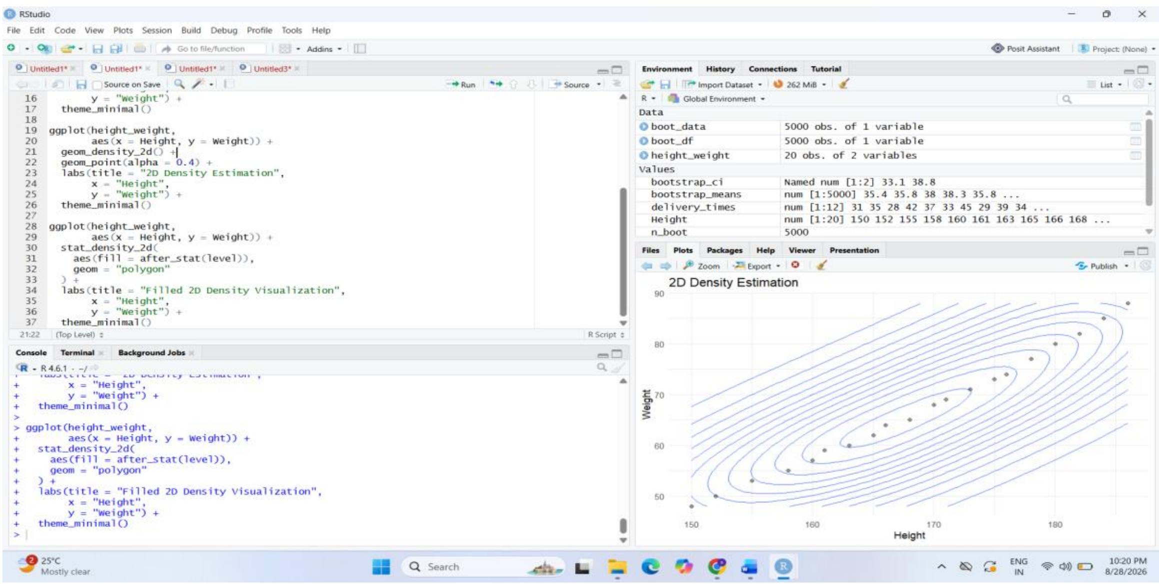
```
# Install package if required
install.packages("ggplot2")

library(ggplot2)

# Given data
Height <- c(150,152,155,158,160,161,163,165,166,168,
        170,171,173,175,176,178,180,182,184,186)

Weight <- c(48,50,53,55,57,59,60,62,64,65,
        68,69,71,73,74,77,80,82,85,88)

height_weight <- data.frame(Height, Weight)

# 2D Histogram
ggplot(height_weight,
     aes(x = Height, y = Weight)) +
  geom_bin2d(bins = 6) +
  labs(title = "2D Histogram of Height and Weight",
     x = "Height",
     y = "Weight") +
  theme_minimal()

# 2D Density Plot
ggplot(height_weight,
     aes(x = Height, y = Weight)) +
```

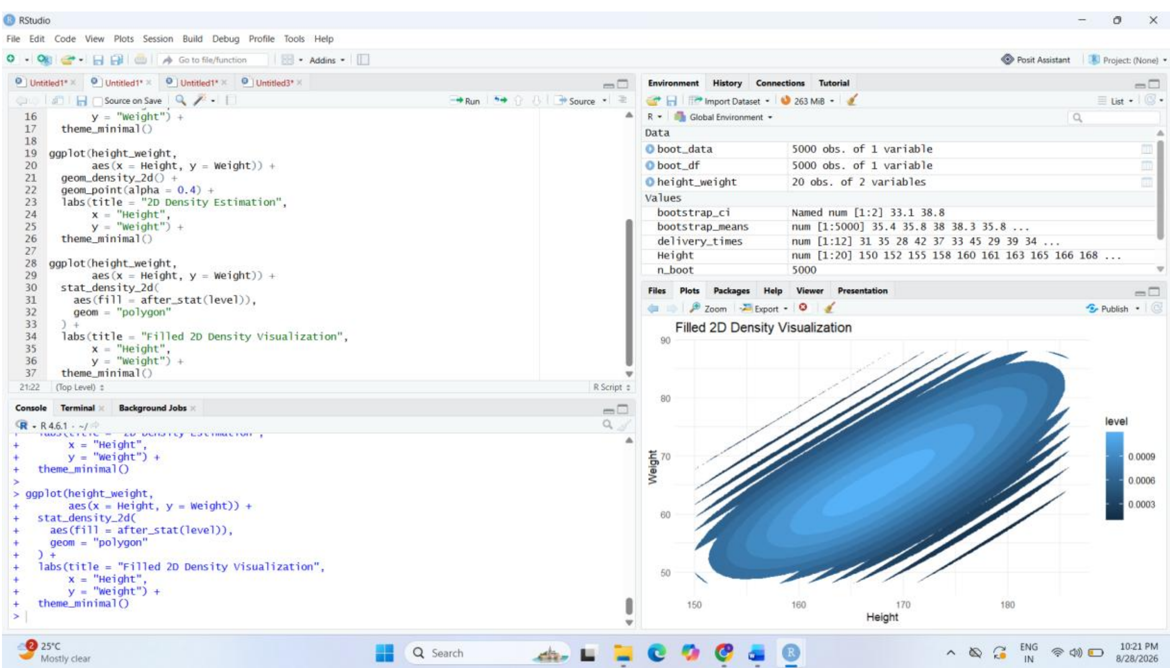
```
 geom_density_2d() +
 geom_point(alpha = 0.4) +
 labs(title = "2D Density Estimation",
     x = "Height",
     y = "Weight") +
 theme_minimal()


# Filled 2D Density Plot
ggplot(height_weight,
     aes(x = Height, y = Weight)) +
 stat_density_2d(aes(fill = after_stat(level)),
               geom = "polygon") +
 labs(title = "Filled 2D Density Visualization",
     x = "Height",
     y = "Weight") +
 theme_minimal()
```

Interpretation:
The 2D histogram divides the height-weight space into rectangular bins and shows where observations are concentrated. The density visualization provides a smoother representation of concentration and makes the structure of the distribution easier to identify.

For this dataset, the **2D density visualization communicates concentration more clearly** because it provides a smoother representation of the observation density instead of depending on discrete rectangular bins.


2D Histogram of Height and Weight
Height
Weight
count


2D Density Estimation
Height
Weight


Filled 2D Density Visualization
Height
Weight
level

5) Color Scales, Accessibility and Misleading Visualization

The question requires two plots: one using a potentially misleading **nonmonotonic/diverging color scheme**, and a redesigned plot using an **ordered, color-vision-deficiency-friendly sequential scale**.

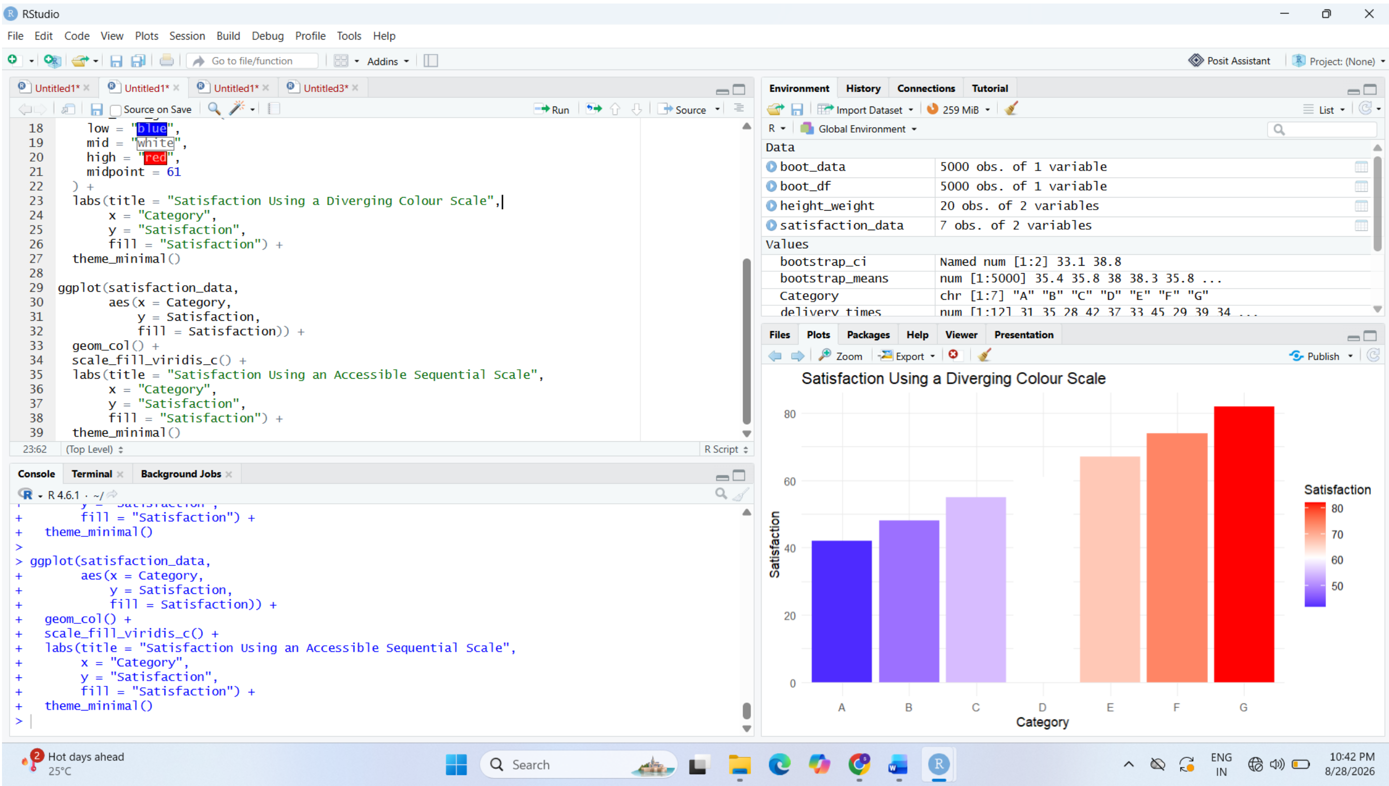
```
# Install packages if required

install.packages("ggplot2")


library(ggplot2)


# Given data

Category <- c("A", "B", "C", "D", "E", "F", "G")

Satisfaction <- c(42, 48, 55, 61, 67, 74, 82)


satisfaction_data <- data.frame(Category,
                                Satisfaction)


# Plot 1: Diverging/nonmonotonic colour scheme

ggplot(satisfaction_data,
       aes(x = Category,
           y = Satisfaction,
           fill = Satisfaction)) +
  geom_col() +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 61
```

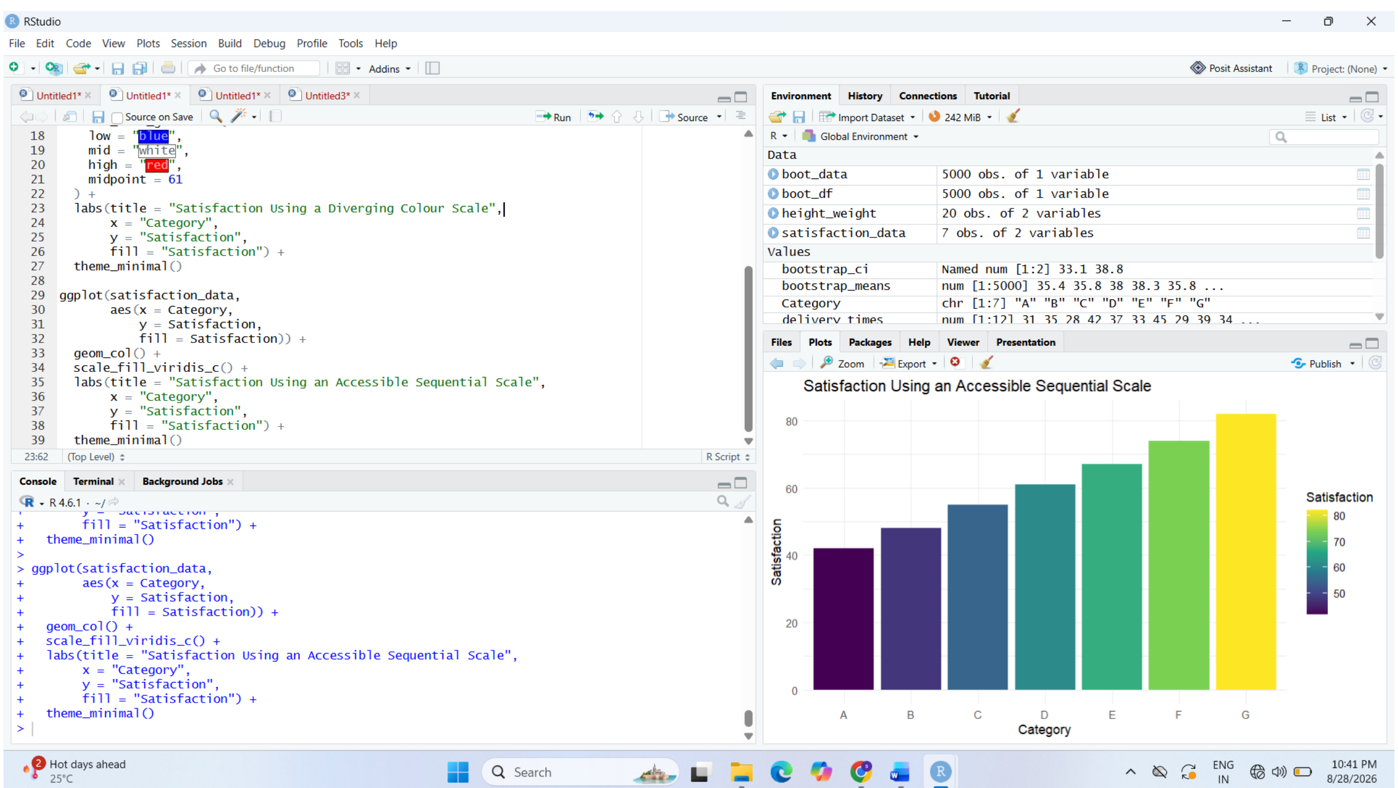
```
 ) +
 labs(title = "Satisfaction Using a Diverging Colour Scale",
     x = "Category",
     y = "Satisfaction",
     fill = "Satisfaction") +
 theme_minimal()


# Plot 2: Ordered sequential colour scale
ggplot(satisfaction_data,
     aes(x = Category,
        y = Satisfaction,
        fill = Satisfaction)) +
 geom_col() +
 scale_fill_viridis_c(option = "viridis") +
 labs(title = "Satisfaction Using an Accessible Sequential Scale",
     x = "Category",
     y = "Satisfaction",
     fill = "Satisfaction") +
 theme_minimal()
```

Interpretation:
The first visualization uses a diverging color scheme. Since the data represent an ordered satisfaction scale from low to high, introducing a midpoint and contrasting colors can imply an artificial division in the data and may distract from the actual increasing pattern.

The second visualization uses an ordered sequential scale. The color intensity changes consistently as satisfaction increases, making the numerical ordering easier to understand. The sequential scale is also more suitable for accessibility and reduces dependence on problematic color distinctions.

Visualization pitfall in the first plot:
A diverging/nonmonotonic color scale can create a false visual emphasis around the selected midpoint and may suggest meaningful categories or boundaries that do not actually exist in the data.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
low = "blue",
mid = "white",
high = "red",
midpoint = 61
) +
labs(title = "Satisfaction Using a Diverging Colour Scale",
x = "Category",
y = "Satisfaction",
fill = "Satisfaction") +
theme_minimal()
ggplot(satisfaction_data,
aes(x = Category,
y = Satisfaction,
fill = Satisfaction)) +
geom_col() +
scale_fill_viridis_c() +
labs(title = "Satisfaction Using an Accessible Sequential Scale",
x = "Category",
y = "Satisfaction",
fill = "Satisfaction") +
theme_minimal()
Environment History Connections Tutorial
Data
boot_data 5000 obs. of 1 variable
boot_df 5000 obs. of 1 variable
height_weight 20 obs. of 2 variables
satisfaction_data 7 obs. of 2 variables
Values
bootstrap_ci Named num [1:2] 33.1 38.8
bootstrap_means num [1:5000] 35.4 35.8 38 38.3 35.8 ...
Category chr [1:7] "A" "B" "C" "D" "E" "F" "G"
Satisfaction Using a Diverging Colour Scale
Satisfaction
Category
Hot days ahead
25°C
10:42 PM
8/28/2026


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
low = "blue",
mid = "white",
high = "red",
midpoint = 61
) +
labs(title = "Satisfaction Using a Diverging Colour Scale",
x = "Category",
y = "Satisfaction",
fill = "Satisfaction") +
theme_minimal()
ggplot(satisfaction_data,
aes(x = Category,
y = Satisfaction,
fill = Satisfaction)) +
geom_col() +
scale_fill_viridis_c() +
labs(title = "Satisfaction Using an Accessible Sequential Scale",
x = "Category",
y = "Satisfaction",
fill = "Satisfaction") +
theme_minimal()
Environment History Connections Tutorial
Data
boot_data 5000 obs. of 1 variable
boot_df 5000 obs. of 1 variable
height_weight 20 obs. of 2 variables
satisfaction_data 7 obs. of 2 variables
Values
bootstrap_ci Named num [1:2] 33.1 38.8
bootstrap_means num [1:5000] 35.4 35.8 38 38.3 35.8 ...
Category chr [1:7] "A" "B" "C" "D" "E" "F" "G"
Satisfaction Using an Accessible Sequential Scale
Satisfaction
Category
Hot days ahead
25°C
10:41 PM
8/28/2026